

Chapter 22

Brewster's angle and retarded potentials

22.1 Brewster's angle

We continue with reflection and refraction at an interface, assuming $\mu' = \mu$ and $\vec{E}$ polarized in the plane of incidence, for which we found

$$E_0'' = E_0 \frac{\tan(\alpha - \beta)}{\tan(\alpha + \beta)} \quad (22.1)$$

If $\alpha + \beta = \pi/2$, then $\tan(\alpha + \beta) \rightarrow \infty$ and hence $E_0'' = 0$: there is no reflection. In this case $\sin \beta = \cos \alpha$, and Snell's law then gives

$$\frac{n'}{n} = \tan \alpha \equiv \tan \alpha_B \quad (22.2)$$

defining the *Brewster angle* α_B . For incidence at this angle, the reflected wave is linearly polarized perpendicular to the plane of incidence.

22.2 Initial value problem in vacuum

Given initial data $A^\nu(0, \vec{y})$ and $\partial_0 A^\nu(0, \vec{y})$, we want to find $A^\nu(x)$ for future times. The solution is

$$A^\nu(x) = \int (D(t, \vec{x} - \vec{y}) \partial_0 A^\nu(0, \vec{y}) + \partial_0 D(t, \vec{x} - \vec{y}) A^\nu(0, \vec{y})) d^3 y \quad (22.3)$$

where

$$\begin{aligned}
D(t, \vec{x}) &= \frac{1}{(2\pi)^3} \int \frac{d^3 k}{2i\omega} (e^{i k x} - e^{-i k x}) \\
&= \frac{1}{4\pi |\vec{x}|} (\delta(ct - |\vec{x}|) - \delta(ct + |\vec{x}|)) \\
&= \frac{1}{2\pi} \text{sign}(ct) \delta(x^2)
\end{aligned} \tag{22.4}$$

with $\omega = c|\vec{k}|$. In the last form, $x^2 = x^\mu x_\mu$: the sign function reflects causality, and the delta function implements the time lag of propagation at the speed of light. The function D satisfies

$$\square D(t, \vec{x}) = 0 \quad , \quad D(0, \vec{x}) = 0 \quad , \quad \partial_0 D(0, \vec{x}) = \delta^3(\vec{x}) \quad , \quad D(-t, \vec{x}) = -D(t, \vec{x}) \tag{22.5}$$

from which one verifies that (22.3) satisfies $\square A^\nu = 0$ along with the initial conditions.

22.3 Adding sources

We now add sources, still in the gauge $\partial_\mu A^\mu = 0$:

$$\square A^\nu = \frac{4\pi}{c} j^\nu \neq 0 \tag{22.6}$$

We write the solution in terms of a Green's function

$$A^\nu(x) = \frac{1}{c} \int G(x, y) j^\nu(y) d^4 y \tag{22.7}$$

with $G(x, y) = G(x - y)$ by Poincaré symmetry. We require

$$\square G(x) = 4\pi \delta^4(x) \tag{22.8}$$

together with the causality condition

$$G(x) = 0 \quad \text{for all } x^0 < 0 \tag{22.9}$$

These two conditions single out a unique solution, $G \equiv G_{\text{retarded}}$.

22.3.1 Finding the Green's function

Away from the origin, rotational symmetry gives, with $r = |\vec{x}|$,

$$\begin{aligned}\square G &= \left(-\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \frac{1}{r} \frac{\partial^2}{\partial r^2} r \right) G \\ &= \frac{1}{r} \left(-\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \frac{\partial^2}{\partial r^2} \right) (r G) = 0 \quad \text{for } r \neq 0\end{aligned}\quad (22.10)$$

The general solution is

$$G(t, r) = \frac{1}{r} (f(ct - r) + g(ct + r)) \quad (22.11)$$

The causality boundary condition eliminates the incoming piece g , leaving

$$G = \frac{1}{r} f(ct - r) \quad (22.12)$$

Now add the point source:

$$\frac{1}{c^2} \frac{\partial^2 G}{\partial t^2} - \nabla^2 G = 4\pi \delta^3(\vec{x}) \delta(t) \quad (22.13)$$

Acting on (22.12), the time derivatives give f''/r , while

$$\begin{aligned}-\nabla^2 \left(\frac{1}{r} f \right) &= -\nabla^2 \left(\frac{1}{r} \right) f - \frac{1}{r} \nabla^2 f - 2 \vec{\nabla} \left(\frac{1}{r} \right) \cdot \vec{\nabla} f \\ &= 4\pi \delta^3(\vec{x}) f - \frac{1}{r} \left(f'' - \frac{2}{r} f' \right) - \frac{2}{r^2} f'\end{aligned}\quad (22.14)$$

using $\nabla^2 f = (1/r) \partial_r^2 (r f) = f'' - (2/r) f'$, $\vec{\nabla}(1/r) = -\vec{x}/r^3$, and $\vec{\nabla} f = -(\vec{x}/r) f'$. All terms cancel against f''/r except the delta function piece, so (22.13) reduces to $4\pi \delta^3(\vec{x}) f = 4\pi \delta^3(\vec{x}) \delta(t)$, that is

$$f(t) = \delta(t) \Rightarrow G = \frac{1}{r} \delta(ct - r) \quad (22.15)$$

Using

$$\delta(x^2) = \delta(c^2 t^2 - r^2) = \frac{1}{2r} (\delta(ct - r) + \delta(ct + r)) \quad (22.16)$$

we can write this as

$$G = 2 \Theta(x^0) \delta(x^2) \quad (22.17)$$

since $\Theta(t) \delta(ct + r) = 0$ for $r > 0$. Similarly, the advanced solution is

$$G_{\text{advanced}} = 2 \Theta(-x^0) \delta(x^2) \quad (22.18)$$

and comparing with (22.4) we note that

$$4\pi D = G_{\text{retarded}} - G_{\text{advanced}} \quad (22.19)$$

Putting everything together in (22.7),

$$\begin{aligned} A^\nu(x) &= \frac{2}{c} \int \Theta(x^0 - y^0) \delta((x - y)^2) j^\nu(y) d^4 y \\ &= \frac{1}{c} \int \frac{1}{|\vec{x} - \vec{y}|} j^\nu \left(t - \frac{|\vec{x} - \vec{y}|}{c}, \vec{y} \right) d^3 y \end{aligned} \quad (22.20)$$

the *retarded potential*: the source at $\vec{y}$ contributes with a time lag $|\vec{x} - \vec{y}|/c$. As a check, imposing the gauge condition $\partial_\nu A^\nu = 0$ on (22.20) requires current conservation, $\partial_\nu j^\nu = 0$.